Assignment 6

Technical Analysis using R, Visualization Phase

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Course	BDA400 - Data Science Tools and Techniques
Assignment title	Assignment 6, Technical Analysis using R, Visualization Phase
Repository	https://github.com/dipakghimire/BDA400_Assignment6

Project Description

For this assignment, I built an interactive portfolio dashboard using R Shiny. The app downloads historical stock prices from Yahoo Finance with the quantmod package (I added Stooq as a backup source) and plots them with ggplot2 as a candlestick, line, or area chart. In the dashboard, I can change the stock symbol, date range, and time frame (daily, weekly, or monthly).

I added moving averages, RSI, and MACD as indicators that can each be switched on or off. I also implemented a moving average crossover rule (20-day and 50-day) that generates Buy, Sell, and Hold signals, and I annotated these signals on the chart.

Files

DipakGhimire_BDA400_A06.R - R Shiny application